Design Document

Theory and Design of the Berkeley ppKTP Squeezer

Katie Gray [1] [1]Department of Physics, University of California, Berkeley

Abstract

This memo documents the optical design of the Berkeley ppKTP squeezed light source: a continuous-wave, subthreshold, degenerate optical parametric amplifier (OPA) in a bow-tie ring cavity targeting −13 to −15 dB of squeezing at dc. It records the reasoning behind each procurement decision so that choices can be revisited as the experiment develops. Three coupled decisions drive the design. First, grey-tracking damage constraints in ppKTP preclude operation at the Boyd–Kleinman optimum ($\xi^* = 2.84$); we operate instead at $\xi \approx 1.0$–1.6, accepting a 27% reduction in conversion efficiency. A longer crystal partially recovers this penalty by lowering the threshold pump power, motivating $l_X \in \{10, 20\}$ mm rather than the 10.5 mm LIGO precedent. Second, the Hall three-port input–output formalism shows that output-coupler transmissivity T_1 must be chosen jointly with l_X: under a 30 mW pump budget and 5 mrad rms phase noise, the optimal T_1 shifts from 7.6% at $l_X = 5$ mm to 14.7% at $l_X = 20$ mm. Third, ABCD cavity enumeration over off-the-shelf mirror radii, filtered through a six-stage stability and mode-quality pipeline, leaves two mechanically viable radii after clipping constraints: $R_c \in \{34.51, 46.01\}$ mm. Together these yield four candidate geometries, all procured for testing. The configuration $l_X = 10$ mm, $R_c = 46.01$ mm closely matches the aLIGO VOPO squeezer and serves as the primary experimental baseline.

1. BACKGROUND & MOTIVATION

The sensitivity of any optical interferometer is ultimately bounded by the quantum mechanical nature of light. Even when every classical noise source has been eliminated, two irreducible vacuum fluctuations remain: photon-counting (shot) noise, arising from the discreteness of photon arrival, and radiation-pressure noise, arising from momentum kicks imparted by individual photons [1]. Caves showed in 1981 that these two noise sources are conjugate quadratures, and proposed the *squeezed-state technique*: inject vacuum that has been "squeezed" into the otherwise unused dark port of the interferometer, suppressing one quadrature of vacuum noise at the price of amplifying the conjugate quadrature [1, 14].

The Berkeley squeezed light source described in this document is designed to operate as a continuous-wave, subthreshold, degenerate optical parametric amplifier (OPA) based on a periodically poled KTP (ppKTP) crystal in a bow-tie ring cavity, targeting ~ -13 to -15 dB of squeezing at dc, consistent with state-of-the-art laboratory sources [9, 10, 15]. The remainder of this document connects each design choice- crystal length, output coupler transmissivity, beam waist size, cavity geometry- to the single observable that matters: the squeezed quadrature variance $V_s(\omega)$ at the output of the cavity. At each step we compare against the existing LIGO VOPO squeezer design [20] and justify each choice against it.

1.1. Input–Output Theory of the Cavity OPA

The intracavity field of a degenerate OPA is governed by the quantum input–output formalism of Collett and Gardiner [2, 3], which treats each cavity mirror as a beam splitter coupling the intracavity mode $\hat{a}$ to a continuum of external vacuum modes. The input field $\hat{a}_{\text{in}}$ entering a port and the output field $\hat{a}_{\text{out}}$ leaving it satisfy the boundary condition $\hat{a}_{\text{out}}(t) = \sqrt{\kappa_i}\,\hat{a}(t) - \hat{a}_{\text{in}}(t)$, and the intracavity field obeys the quantum Langevin equation

$$\dot{\hat{a}} = -\frac{\kappa}{2}\hat{a} + g\,\hat{a}^\dagger - \sqrt{\kappa_1}\,\hat{a}_{\text{in}}^{(1)} - \sqrt{\kappa_2}\,\hat{a}_{\text{in}}^{(2)} - \sqrt{\kappa_X}\,\hat{a}_{\text{in}}^{(X)}, \quad (1)$$

where g is the parametric gain rate, $\kappa = \kappa_1 + \kappa_2 + \kappa_X$ is the total amplitude decay rate, and $\hat{a}_{\text{in}}^{(1)}$, $\hat{a}_{\text{in}}^{(2)}$, $\hat{a}_{\text{in}}^{(X)}$ are delta-correlated vacuum operators entering through the output coupler (mirror 1), the control mirror (mirror 2), and the crystal loss channel respectively [3, 5].

Solving Eq. (1) in the frequency domain gives the power spectral density of the squeezed (−) and anti-squeezed (+) quadratures,

$$V^{\pm}(\omega) = 1 \pm \eta\,\frac{4x}{(1 \mp x)^2 + (\omega/\kappa)^2}, \quad (2)$$

where $x = g/\kappa$ is the normalised pump coupling ($x = 1$ at threshold), $\eta = \kappa_1/\kappa$ is the escape efficiency, and $\hbar = 1$ normalises the vacuum variance to unity [2, 14, 11]. At dc this reduces to the standard result

$$V_s(0) \simeq 1 - \frac{4\eta x}{(1+x)^2}. \quad (3)$$

The exact dc variance, retaining all three vacuum ports without the $\kappa_i \ll \kappa$ expansion implicit in Eq. (3), is given by the Hall input–output treatment [17]:

$$V_s(0) = \frac{(r_1 - re^{-\Gamma}/r_1)^2 + (t_1 t_2)^2 + (t_1 r_2 r_X)^2}{(1 - re^{-\Gamma})^2}, \quad (4)$$

where $r_1 = \sqrt{1-T_1}$, $r_2 = \sqrt{1-T_2}$, $t_1 = \sqrt{T_1}$, $t_2 = \sqrt{T_2}$, $r = r_1 r_2 t_X$ is the round-trip amplitude reflectivity,

$t_X = e^{-\alpha l_X/2}$ is the single-pass crystal transmission at absorption coefficient α, and $r_X = \sqrt{1-t_X^2}$ is the vacuum coupling amplitude through crystal loss.

Unlike the single-port Collett–Gardiner approximation, which lumps all loss into one effective decay rate, the three-port Hall decomposition resolves the crystal itself into a vacuum noise injection port distinct from the output and control mirrors. Consequently, Eq. (4) reduces to Eq. (3) in the limit $\kappa_1, \kappa_2, \kappa_X \ll \kappa$. At $\alpha = 0.02\,\mathrm{m}^{-1}$ [18] and $T_1 = 10\%$, the crystal-loss rate κ_X contributes 0.09% of κ at $l_X = 5\,\mathrm{mm}$, rising to 0.37% at $l_X = 20\,\mathrm{mm}$: small at every length considered, but growing monotonically with the crystal length used to increase gain, so it cannot be neglected if l_X is considered a free design variable.

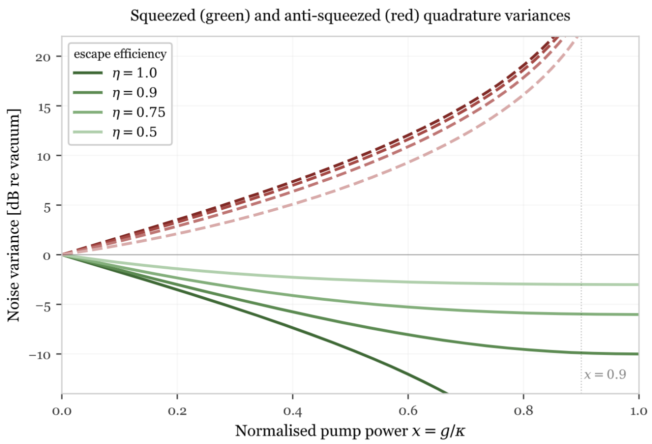


Figure 1: Squeezed (green) and anti-squeezed (red) quadrature variances as a function of normalised pump power $x = g/\kappa$, for several values of the escape efficiency η. As $x \to 1$ the squeezed quadrature asymptotes to the loss floor $1-\eta$, while the anti-squeezed quadrature diverges; the dashed vertical line marks the practical operating point $x = 0.9$.

1.2. Crystal Length and Escape Efficiency

The design parameters typically optimised for an OPA are crystal length l_X, output coupler transmissivity T_1, beam waist w_0, and pump power. Equation (3) demonstrates that each of these enters the optimisation only through its effect on x or η. Crystal length plays a double role: it sets the parametric gain rate $g \propto |\gamma|\, l_X$, which determines x relative to threshold, and it sets the intrinsic loss rate $\kappa_X = \alpha l_X/2\tau_a$, which enters η directly. The transverse dimensions of the crystal (height and width) affect neither gain nor loss in the paraxial regime; they matter only through secondary effects (diffraction and clipping loss, thermal gradients, photorefractive damage) and can be chosen independently once l_X is fixed.

The escape efficiency can also be written as

$$\eta = \frac{\kappa_1}{\kappa} = \frac{\ln r_1}{\ln r} = \frac{T_1/2}{T_1/2 + T_2/2 + \alpha l_X/2} \tag{5}$$

and is the probability that an intracavity photon exits through the useful output port rather than being lost to the control mirror or the crystal. As $x \to 1$, $V_s(0) \to 1-\eta$, so the maximum achievable squeezing is

$$\mathcal{S}_{\max} = -10\log_{10}(1-\eta)\ \mathrm{dB}, \tag{6}$$

set entirely by the ratio of useful output coupling to total cavity loss, independent of pump power, crystal length, or beam waist [2, 14].

Because T_1 enters η directly while l_X enters both g and κ_X at once, the two cannot be optimised independently: the value of T_1 that maximises squeezing at fixed pump budget depends on l_X, and vice versa. Section 4 treats this as the joint two-parameter optimisation it is, with the Boyd–Kleinman waist of Section 2 fixing the remaining degree of freedom in g.

2. BOYD–KLEINMAN FOCUSING

The plane-wave coupled equations governing parametric gain assume infinite transverse beam profiles. In reality, a laser is a focused Gaussian beam of waist w_0 propagating along the crystal axis. For this case, Boyd and Kleinman showed that the second-harmonic (or parametric) conversion power acquires a dimensionless correction factor h [19]:

$$P_{\mathrm{SHG}} = \frac{2\omega_1^2 d_{\mathrm{eff}}^2 l_X k_1}{\pi\varepsilon_0 c^3 n_1^2 n_2} \cdot P_1^2 \cdot h_{\mathrm{SHG}}(\sigma, \xi, B), \tag{7}$$

where P_1 is the fundamental power, $k_1 = n_1\omega_1/c$ is the fundamental wavenumber, and the focusing function is the double integral

$$h(\sigma,\xi,B) = \frac{1}{4\xi}\int_{-\xi}^{+\xi} d\tau_1 \int_{-\xi}^{+\xi} d\tau_2\, \frac{e^{i\sigma(\tau_1-\tau_2)}\cdot e^{-B^2(\tau_1-\tau_2)^2/\xi}}{(1+i\tau_1)(1-i\tau_2)}. \tag{8}$$

The arguments encode the three physical effects competing inside the crystal. The *focusing parameter* $\xi = l_X/b$, where $b = w_0^2 k_1$ is the confocal parameter, measures how many Rayleigh lengths fit inside the crystal: $\xi \ll 1$ is loose focusing (long z_R, gentle divergence) and $\xi \gg 1$ is tight focusing (short z_R, rapid divergence). The phase-mismatch parameter $\sigma = b\Delta k/2$ measures the accumulated phase walk-off scaled by the confocal length. Jointly maximising h over both ξ and σ gives the Boyd–Kleinman optimum

$$\xi^* = 2.84, \qquad \sigma^* \approx 0.573, \qquad h^* = 1.068, \tag{9}$$

the theoretical ideal operating point for the OPA [19].

2.1. Optimal Beam Waist as a Function of Crystal Length

The optimal focusing parameter fixes the pump beam waist through $\xi = l_X/b = l_X\lambda/(2\pi n w_0^2)$:

$$w_0^*(\xi) = \left(\frac{l_X \lambda_{\mathrm{pump}}}{2\pi n_{\mathrm{crystal}}\,\xi}\right)^{1/2}, \tag{10}$$

where $\lambda_{\mathrm{pump}} = 532\,\mathrm{nm}$ and $n_{\mathrm{crystal}} \approx 1.889$ at 532 nm. At $\xi = \xi^*$, $l_X = 10\,\mathrm{mm}$ gives $w_0^* \approx 12.9\,\mu\mathrm{m}$, while $l_X =$

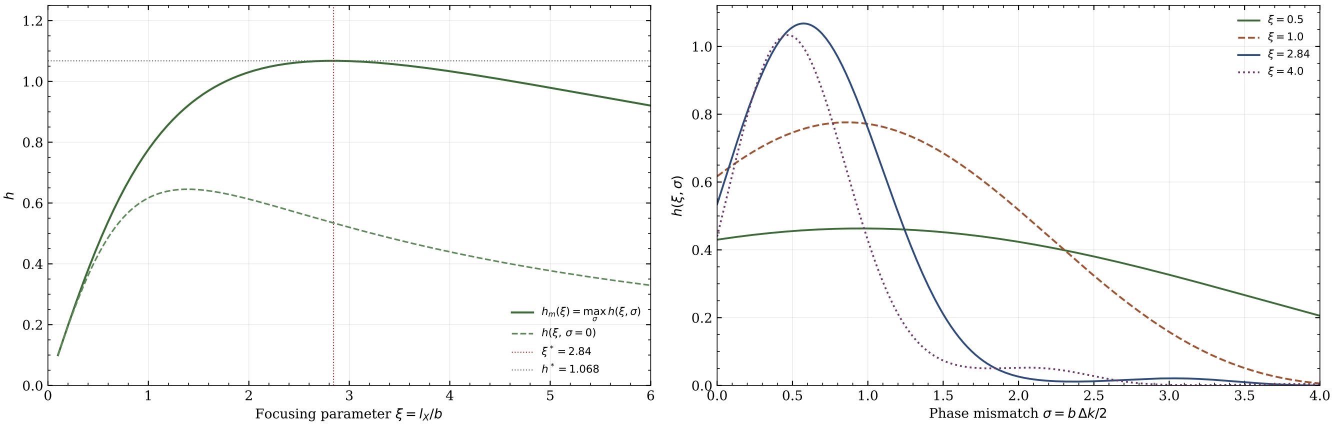


Figure 2: Two-dimensional slices of $h(\sigma, \xi)$. *Left:* $h_m(\xi) \equiv \max_\sigma h(\sigma, \xi)$ (solid green) peaks at $\xi^* = 2.84$, $h^* = 1.068$ (dotted lines), while $h(\xi, \sigma = 0)$ (dashed green) peaks spuriously at $\xi \approx 1.4$. *Right:* $h(\xi, \sigma)$ versus σ for four values of ξ. Tighter focusing ($\xi = 4.0$) demands precise phase matching and has a narrow acceptance bandwidth; loose focusing ($\xi = 0.5$) is very forgiving. The $\xi = 2.84$ curve achieves the highest peak while retaining a reasonable bandwidth.

20 mm shifts the optimum to $w_0^* \approx 17.8\,\mu$m, scaling as $\sqrt{l_X}$ (Fig. 3). The LIGO OPA cavity instead operates at $w_0 = 21.1\,\mu$m for a 10 mm crystal, corresponding to $\xi \approx 1$ [20], a deliberate departure from the BK optimum, for reasons we will quantify.

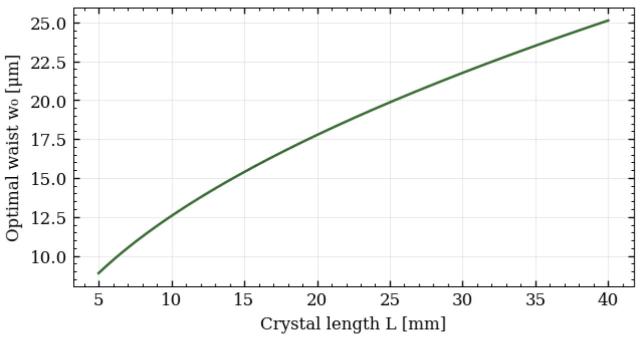


Figure 3: Optimal 532 nm pump beam waist w_0^* versus crystal length l_X from Eq. (10) at $\xi = 2.84$.

2.2. Operating Off-Maximum: Thermal and Damage Considerations

The peak of $h_m(\xi)$ at $\xi^* = 2.84$ is shallow, but the cost of operating away from it is worth quoting precisely. At $\xi = 1$ the optimal σ itself shifts, from $\sigma^* \approx 0.573$ at ξ^* to $\sigma_{\text{opt}}(\xi{=}1) \approx 0.861$; re-optimising σ at fixed $\xi = 1$ gives

$$h(1,\,\sigma_{\text{opt}}) = 0.776, \qquad \frac{h(1, \sigma_{\text{opt}})}{h^*} = 0.727, \tag{11}$$

a 27% reduction in conversion efficiency relative to the true optimum.

While operating at the BK optimum is ideal, it is often ruled out by damage constraints. KTP exhibits photochromic grey-tracking, in which prolonged exposure to intense 532 nm radiation degrades the refractive index and increases intracavity loss [23, 22]. The peak circulating intensity at the crystal face is

$$I_{\text{peak}} = \frac{2P_{\text{circ}}}{\pi w_0^2}, \tag{12}$$

and at the BK-optimal waist $w_0^* \approx 12.9\,\mu$m with $P_{\text{circ}} = 30$ mW this gives $I_{\text{peak}} \approx 1.1 \times 10^6$ W/cm^2, roughly an order of magnitude above the practical grey-tracking threshold of $\lesssim 10^5$ W/cm^2 reported for ppKTP [23, 22]. Operating instead at $\xi = 1$ ($w_0 \approx 21\,\mu$m) reduces this to $I_{\text{peak}} \approx 4.3 \times 10^4$ W/cm^2, comfortably below the damage threshold, at the cost of the 27% efficiency penalty quantified above.

A longer crystal relaxes this trade-off on two fronts simultaneously: it lowers the threshold pump power *and* shrinks the penalty for operating away from tight BK focus, making it possible to choose a thermally safe waist without sacrificing much circulating power. This is the central reason this design favours a longer crystal than the LIGO precedent.

3. PHASE MATCHING AND TEMPERATURE STABILITY

The ppKTP crystal must satisfy the quasi-phase-matching condition $\Delta k_{\text{eff}} = 0$ to sustain efficient parametric down-conversion. For a periodically poled crystal of poling period Λ,

$$\Delta k_{\text{eff}} = k_p - k_s - k_i - \frac{2\pi}{\Lambda} = 0, \tag{13}$$

where Λ is fixed at fabrication to satisfy this condition at a design temperature $T_0 = 35$°C. All three refractive indices drift with temperature, so Δk_{eff} acquires a temperature dependence $\Delta k(T)$ away from T_0, and the conversion efficiency falls off as

$$\eta_{\text{PM}}(T) = \text{sinc}^2\!\left(\frac{\Delta k(T)\, l_X}{2}\right), \tag{14}$$

with $\Delta k(T)$ evaluated from the temperature-dependent KTP Sellmeier equations of Ref. [21]. The efficiency

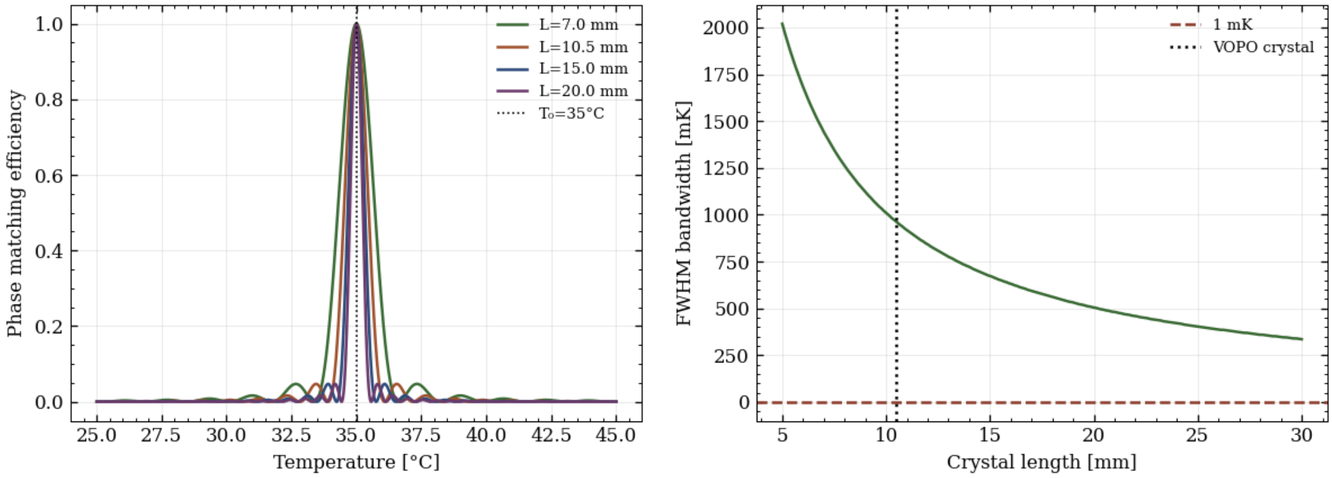


Figure 4: *Left:* Phase-matching efficiency $\eta_{\mathrm{PM}}(T)$ for four crystal lengths, centred on $T_0 = 35°\mathrm{C}$; bandwidth narrows as $1/l_X$. *Right:* FWHM thermal acceptance bandwidth ΔT vs. crystal length l_X. The 1 mK controller floor (red dashed) lies orders of magnitude below the acceptance bandwidth for all lengths considered.

drops to half its peak value at $\Delta k\, l_X/2 = \pm 0.4429\pi$, giving a FWHM thermal acceptance bandwidth

$$\Delta T_{\mathrm{FWHM}} \approx \frac{1.772\pi}{l_X\, |d\Delta k/dT|}, \tag{15}$$

with $d\Delta k/dT$ evaluated numerically by finite difference on $\Delta k(T)$ at T_0. Because l_X appears only in the denominator, ΔT_{FWHM} scales as $1/l_X$: a longer crystal demands phase agreement across a longer propagation distance, and so tolerates a narrower range of temperatures, exactly as a longer aperture or a longer time window narrows the resolved bandwidth in any Fourier-conjugate pair.

Figure 4 (left) confirms this directly: all four crystal lengths peak at the same $T_0 = 35°\mathrm{C}$, since the poling period compensates the phase mismatch at the design point regardless of length, while the peak narrows sharply as l_X increases. The bandwidth falls from $1.01°\mathrm{C}$ at $l_X = 10\,\mathrm{mm}$ to $337\,\mathrm{mK}$ at $l_X = 30\,\mathrm{mm}$ (Fig. 4, right), which are both values orders of magnitude above the 1 mK stability floor of a standard PID temperature controller. Temperature control is therefore not a limiting constraint on the crystal length choice over $l_X = 5$–$30\,\mathrm{mm}$, and l_X can be selected freely on the basis of the gain and threshold arguments of Sections 2 and 4.

4. SQUEEZING OPTIMISATION VIA THE HALL FORMALISM

With the Boyd–Kleinman waist of Section 2 fixing $w_0(l_X, \xi)$, we now fold this into the full cavity squeezing model. The single-pass parametric gain corrected for focused Gaussian modes is

$$\Gamma^2 = \frac{\omega_a^2 d_{\mathrm{eff}}^2\, l_X^2}{\varepsilon_0 c^3 n_a^2 n_b} \cdot \frac{P_{\mathrm{circ}}}{A_{\mathrm{eff}}} \cdot h(\xi, \sigma), \tag{16}$$

where P_{circ} is the circulating pump power and $A_{\mathrm{eff}} = \pi w_0^2/2$ is the effective Gaussian mode area, with the factor of $\frac{1}{2}$ relative to the geometric spot area arising from the peak intensity of a Gaussian beam, $I_{\mathrm{peak}} = 2P/\pi w_0^2$. Substituting the Boyd–Kleinman waist $w_0^2 = l_X \lambda_b/(2\pi n_b \xi)$ from Eq. (10) fixes the l_X dependence of Γ^2 explicitly:

$$\Gamma^2 = \frac{\omega_a^2 d_{\mathrm{eff}}^2 l_X^2}{\varepsilon_0 c^3 n_a^2 n_b} \cdot \frac{2P_{\mathrm{circ}}}{\pi w_0^2} \cdot h = \frac{4\,\omega_a^2 d_{\mathrm{eff}}^2}{\varepsilon_0 c^3 n_a^2} \cdot \xi\, h(\xi, \sigma)\, P_{\mathrm{circ}}\, l_X, \tag{17}$$

i.e. $\Gamma^2 \propto \xi\, h(\xi, \sigma)\, P_{\mathrm{circ}}\, l_X$: the gain grows *linearly* with crystal length at fixed ξ, since the waist itself grows as $\sqrt{l_X}$ to hold ξ fixed, and the l_X^2 in the numerator only partially cancels the $1/A_{\mathrm{eff}} \propto 1/l_X$ from the correspondingly larger mode area. This is the reason a longer crystal lowers the threshold pump power despite its larger intrinsic loss.

The threshold single-pass gain is $\Gamma^* = -\ln r$, where $r = r_1 r_2 t_X$ is the round-trip amplitude, and the normalised coupling is $x = \Gamma/\Gamma^*$. Squeezing performance therefore depends on four quantities: the gain $\Gamma(l_X, w_0, P_{\mathrm{circ}})$ set by Eq. (17); the threshold $\Gamma^*(T_1, T_2, \alpha l_X)$ set by mirror transmissivities and crystal loss; the escape efficiency $\eta(T_1, T_2, \alpha l_X)$; and the rms phase noise ϕ_{rms}, which mixes the anti-squeezed quadrature into the squeezed one.

Figure 5, makes both halves of this trade-off visible together: threshold power drops by roughly an order of magnitude between $l_X = 5\,\mathrm{mm}$ and $l_X = 20\,\mathrm{mm}$ at any fixed T_1 (top), while the squeezing floor at the same two lengths degrades by less than 1 dB (bottom). Fig. 5 (middle) shows for $T_1 = 10\%$, $T_2 = 0.2\%$, $\eta \approx 0.978$–0.980 across $l_X = 5$–$20\,\mathrm{mm}$, corresponding to a squeezing floor of -16.5 to $-17.1\,\mathrm{dB}$.

We now see that the crystal-length choice is once again lopsided in favour of longer crystals, as the threshold-power benefit is large and the floor penalty is small provided the beam waist can be kept inside the thermal and damage limits discussed in Section 2.2.

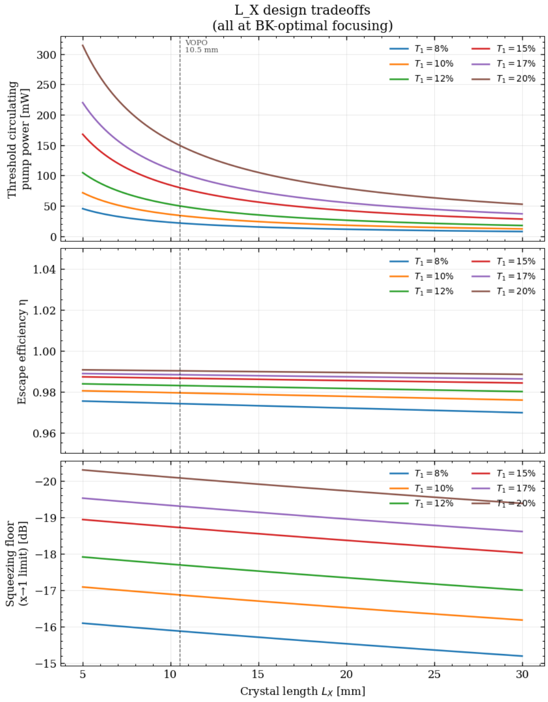


Figure 5: Crystal-length trade-offs at BK-optimal focusing, for six values of T_1. *Top:* threshold pump power vs. l_X, from Eq. (17). *Centre:* escape efficiency η, set almost entirely by T_1. *Bottom:* squeezing floor vs. l_X. Dashed line marks the 10.5 mm LIGO reference crystal.

4.1. Parameter Sweep: Output Coupler Transmissivity

Figure 6 shows dc squeezing, escape efficiency, and threshold power against T_1 for four crystal lengths at fixed $x = 0.8$. At low T_1 the escape efficiency $\eta \ll 1$ and squeezing is poor regardless of pump power, since most intracavity photons are lost to κ_2 and κ_X rather than extracted through the output coupler; as T_1 increases, $\eta \to 1$ and squeezing improves, but threshold pump power rises steeply since $\Gamma^* \approx T_1/2 + T_2/2 + \alpha l_X/2$ grows with T_1. At fixed T_1, shorter crystals give more squeezing, reflecting their smaller intracavity loss fraction, but Eq. (17) means longer crystals reach a given x at dramatically lower pump power, so the comparison at fixed T_1 understates their advantage at fixed pump budget.

At a fixed pump budget rather than fixed x, this trade-off sharpens further: squeezing is monotonically decreasing in T_1 once η has saturated, so there is no interior optimum in T_1. In other words, the best achievable squeezing at any crystal length is found immediately next to the $x = 0.95$ instability boundary, at the *lowest* T_1 the budget can sustain without the design becoming unstable. The operating point is therefore set by how close to threshold the design is willing to run, which is exactly where phase noise enters.

4.2. Phase Noise

A phase fluctuation $\delta\phi$ in the squeezing angle rotates the noise ellipse, mixing the amplified quadrature into the squeezed one. The exact observed variance under a pure quadrature rotation is

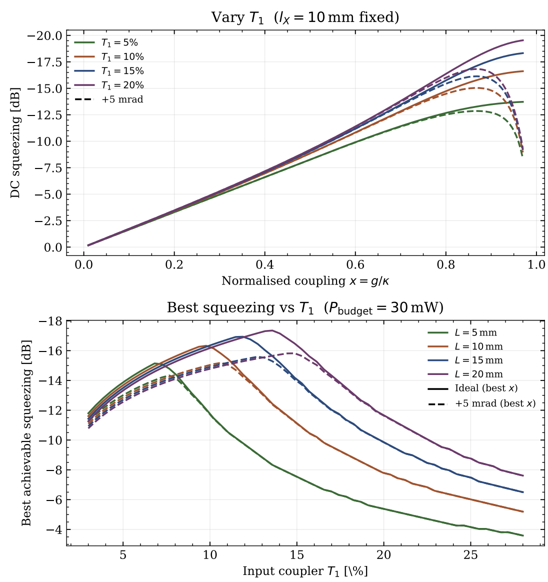
$$V_s^{\rm obs} = V_s \cos^2\delta\phi + V_c \sin^2\delta\phi, \tag{18}$$

where $V_c = 1+4\eta x/(1-x)^2$ is the anti-squeezed variance (Eq. (2) at dc). To leading order in $\delta\phi^2_{\rm rms}$ this gives $V_s^{\rm obs} \approx V_s + (V_c - V_s)\delta\phi^2_{\rm rms}$: the penalty scales with $V_c - V_s \approx V_c$ near threshold, where $V_c \gg 1$, so even small jitter ($\sim$5 mrad) erases much of the squeezing benefit of pushing x close to 1.

Table 1: Phase-noise-limited optimum T_1 and best achievable dc squeezing versus crystal length, at fixed 30 mW circulating pump budget and 5 mrad rms phase jitter, compared against the ideal (noise-free) optimum at the same budget.

	Ideal		*+5 mrad noise*	
l_X [mm]	$T_1^{\rm opt}$ [%]	best [dB]	$T_1^{\rm opt}$ [%]	best [dB]
5.0	6.9	−15.2	7.6	−14.3
10.0	9.6	−16.4	10.6	−15.2
15.0	11.5	−17.0	12.9	−15.6
20.0	13.2	−17.4	14.7	−15.9

Figure 7: DC squeezing under phase noise vs x at fixed $l_X = 10$ mm for varying T_1 (top), and best achievable squeezing vs T_1 at fixed pump budget varying l_X (bottom).



This reshapes the optimal T_1 rather than simply imposing a flat ceiling. The true optimum is a balance between two competing pulls: lower T_1 reduces threshold pump power (since $\Gamma^* \approx T_1/2 + T_2/2 + \alpha l_X/2$ grows with T_1), allowing a given x to be reached with less circulating power; while higher T_1 improves the escape efficiency η and therefore the squeezing ceiling $\mathcal{S}_{\max} = -10\log_{10}(1-\eta)$ dB. At a fixed pump budget, once T_1

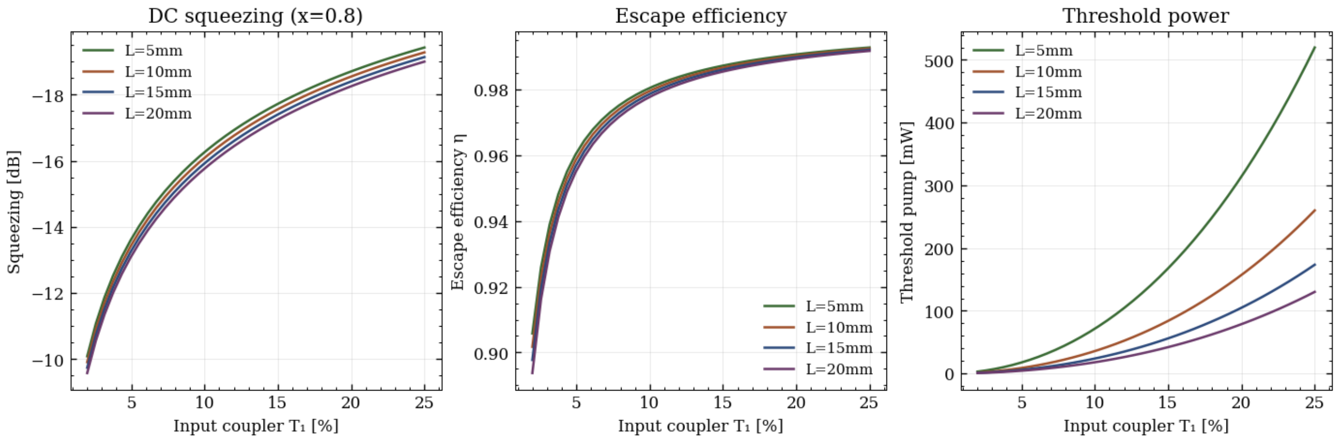


Figure 6: DC squeezing $V_s(0)$ (left), escape efficiency η (centre), and threshold pump power (right) versus output coupler transmissivity T_1, at fixed $x = 0.8$ and $\alpha = 0.02\,\mathrm{m}^{-1}$, for four crystal lengths.

is large enough that the threshold exceeds the budget, squeezing collapses regardless of η. Because $\Gamma^2 \propto l_X$ (Eq. (17)) lowers the threshold for longer crystals, a longer crystal tolerates a higher T_1 before this budget constraint bites, shifting the balance toward higher escape efficiency. Maximising V_s^{obs} over x at each T_1, for a fixed 30 mW pump budget and 5 mrad rms phase jitter, gives the optimum tabulated in Table 1 alongside its noise-free counterpart, and shown against x and against T_1 in Fig. 7.

The shift between the ideal and noisy columns is itself informative: phase noise moves the optimum T_1 *up* relative to the ideal case at every length (by roughly 0.7–1.5 percentage points), because a slightly higher T_1 trades a small amount of escape efficiency for a lower x at the same squeezing, pulling the operating point back from the threshold region where V_c diverges fastest. The practical conclusion is the opposite of a single fixed T_1 recommendation: *the phase-noise-optimal* T_1 *should be chosen jointly with* l_X, not fixed at a round number such as 10% independent of crystal length. For a 20 mm crystal, the phase-noise-limited optimum sits at $T_1 \approx 14.7\%$, not the lower value that would be inferred from a shorter-crystal reference design.

5. ABCD CAVITY GEOMETRY

Having established the squeezing performance as a function of crystal and mirror parameters, we now turn to the geometric problem of designing a bow-tie cavity that (i) supports the target intracavity waist w_0^* at the crystal, (ii) remains stable at both 1064 nm and 532 nm simultaneously, (iii) keeps astigmatism below a tolerable level, (iv) avoids transverse-mode degeneracies near the carrier resonance, and (v) can physically be built without the beam clipping on the mirror mounts. The first four constraints are evaluated computationally by the six-stage optimizer of Section 5.2. The fifth lies outside the ABCD ray-optics model entirely, since it depends on mount hardware rather than beam propagation; it is addressed separately, in Section 6. Section 5.1 below describes the computational method in full before any of it is applied; Section 6 then reports the results.

5.1. ABCD Matrix Propagation and Cavity Stability

For a bow-tie cavity with two identical curved mirrors (radius of curvature R_c, separated by length d_1), two flat mirrors, a flat section of total path length d_2 between the flat mirrors, and a crystal of physical length l_X and refractive index n_a at signal wavelength λ_a, the round-trip ray transfer matrix M is the ordered product of eleven individual ABCD matrices evaluated starting from the crystal centre [15].

The constituent matrices are free-space propagation $M_{\mathrm{prop}}(d)$, the air–crystal interface $M_{\mathrm{int}}(n_1, n_2)$, and the curved mirror in tangential and sagittal planes respectively,

$$M_{\mathrm{prop}}(d) = \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix} \qquad M_{\mathrm{int}}(n_1, n_2) = \begin{pmatrix} 1 & 0 \\ 0 & n_1/n_2 \end{pmatrix}$$

$$M_{\mathrm{curv}}^{(t)} = \begin{pmatrix} 1 & 0 \\ -2\cos\theta/R_c & 1 \end{pmatrix} \qquad M_{\mathrm{curv}}^{(s)} = \begin{pmatrix} 1 & 0 \\ -2/(R_c\cos\theta) & 1 \end{pmatrix}$$

The crossing half-angle θ distinguishes the two planes: tighter crossing ($\theta \to 0$) requires a mechanically inconvenient geometry where alignment and beam clipping push the design away from being robust, while larger θ introduces increasing astigmatism. θ is therefore treated as a free parameter of the search rather than fixed in advance.

The round-trip propagates from the crystal centre, through the half-crystal (using the reduced optical path $l_X/2n_a$ in ABCD units), exits via the air–crystal interface, traverses the free-space gap $(d_1 - l_X)/2$ to the first curved mirror, reflects, traverses the diagonal arm d_3 to the first flat mirror, reflects, propagates through the flat section d_2, reflects off the second flat mirror, traverses the diagonal arm d_3 back to the second curved mirror, reflects, and returns symmetrically.

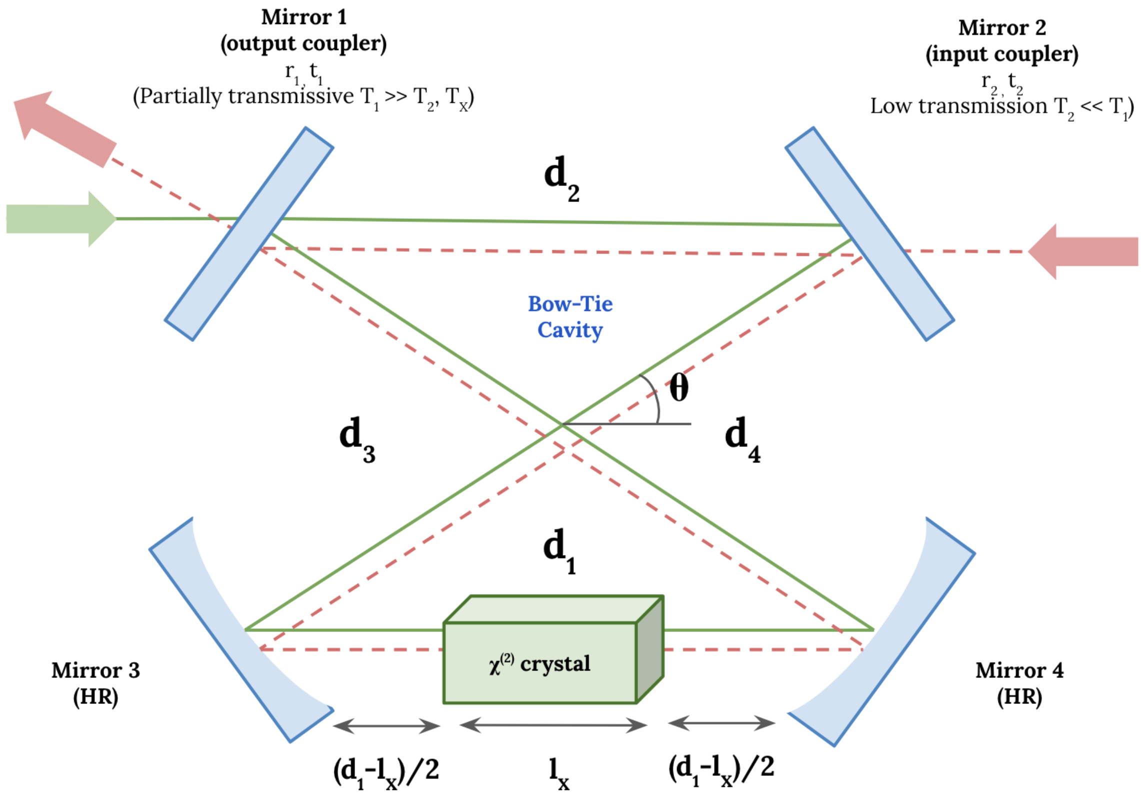


Figure 8: Schematic of the bow-tie OPA cavity. Mirrors 3 and 4 (bottom, curved, HR) are separated by d_1, with the $\chi^{(2)}$ crystal centred between them leaving gaps of $(d_1 - l_x)/2$ on each side. The two diagonal crossing arms of length d_3 meet at half-angle θ, connecting to the flat top section d_2 between the output coupler (Mirror 1, $T_1 \gg T_2$) and input coupler (Mirror 2, $T_2 \ll T_1$). The green and red beams indicate the 532 nm pump and 1064 nm signal paths respectively.

$$M = M_{\text{half}}\, M_{\text{int}}^{a\to x}\, M_{\text{gap}}\, M_{\text{curv}}\, M_{d_3}\, M_{\text{flat}}\, M_{d_2} \\ \times M_{\text{flat}}\, M_{d_3}\, M_{\text{curv}}\, M_{\text{gap}}\, M_{\text{int}}^{x\to a}\, M_{\text{half}}, \tag{19}$$

where matrices act right-to-left, $M_{\text{int}}^{a\to x}$ and $M_{\text{int}}^{x\to a}$ denote the air–crystal and crystal–air interfaces respectively, $M_{\text{half}} = M_{\text{prop}}(l_X/2n_a)$ is the half-crystal propagation, and $M_{\text{flat}} = I$ for ideal flat mirrors.

The diagonal arm length is determined by the geometric closure condition

$$d_3 = \frac{d_1 + d_2}{2\cos\theta}, \tag{20}$$

which eliminates d_3 as a free parameter, leaving $\{R_c, d_1, d_2\}$ as the three independent variables; imposing the target waist then fixes one further combination of these. The stability condition and waist follow from the round-trip matrix elements (A, B, D):

$$m = \frac{A + D}{2}, \qquad |m| < 1, \tag{21}$$

$$w_0 = \sqrt{\frac{\lambda |B|}{\pi\sqrt{1 - m^2}}}. \tag{22}$$

Equation (22) is evaluated independently in the tangential and sagittal planes, giving $w_{0,t}$ and $w_{0,s}$; the effective waist used to compute ξ is the geometric mean $w_{0,\text{eff}} = \sqrt{w_{0,t}\, w_{0,s}}$.

The robust cavity design being targeted must be stable with and without the crystal, which is important for diagnostics and initial alignment before the crystal is inserted.

5.2. Six-Stage Filter Pipeline

The search space (R_c, d_1, d_2) is scanned over a set of candidate radii of curvature under consideration, with d_1 and d_2 stepped finely over physically meaningful ranges. For each candidate geometry, six sequential filters are applied, ordered by computational cost: cheap geometric checks run first so that the expensive ABCD eigenmode and transverse-mode calculations are only performed on candidates that have already passed the basic sanity checks.

Stage 1 (geometry & empty-cavity stability). Compute d_3 from Eq. (20); reject if $d_3 \le 0$. Before any wavelength-dependent ABCD calculation is attempted, the cavity is also checked for stability with the crystal replaced by air, in both the tangential and sagittal planes. Any candidate that fails this empty-cavity check is rejected immediately, avoiding the cost of the five subsequent stages.

Stage 2 (green waist). The 532 nm round-trip matrix is computed in both the tangential and sagittal planes. A candidate is rejected if either plane is individually unstable. For the two stable waists $w_{0,t}^{(g)}$ and $w_{0,s}^{(g)}$, the effective waist is taken as the geometric mean $w_{0,\mathrm{eff}}^{(g)} = \sqrt{w_{0,t}^{(g)}\, w_{0,s}^{(g)}}$, which approximates the waist that the pump mode volume samples when computing the Boyd–Kleinman overlap integral. The candidate is accepted at this stage only if $w_{0,\mathrm{eff}}^{(g)}$ falls within a target window $[w_0(\xi_{\max}),\, w_0(\xi_{\min})]$ set by the chosen Boyd–Kleinman range, Eq. (23). This is the strictest optical filter in the pipeline because the green waist directly sets the nonlinear coupling strength Γ; an error here cannot be corrected by any downstream optic.

$$\frac{|w_{0,\mathrm{eff}}^{(g)} - w_{0,\mathrm{target}}^{(g)}|}{w_{0,\mathrm{target}}^{(g)}} < 5\%, \tag{23}$$

Stage 3 (astigmatism, green). Compute the fractional astigmatism

$$\mathcal{A} = \frac{|w_{0,t} - w_{0,s}|}{w_{0,\mathrm{eff}}}; \tag{24}$$

reject if $\mathcal{A} > 10\%$ in the green mode.

Stage 4 (red waist & astigmatism). The 1064 nm round-trip matrix is computed in both planes and the candidate is rejected if either is unstable, or if the red astigmatism exceeds the same 10% bound. No tight waist-matching tolerance is imposed on the red mode: the green waist sets the nonlinear coupling, whereas the red waist determines the signal mode that must be matched into the homodyne detector, and mode-matching optics after the cavity output can compensate for moderate red waist errors.

Stage 5 (sensitivity & cavity metrics). The sensitivity of the Boyd–Kleinman focusing parameter ξ to errors in the curved-mirror separation d_1 is evaluated by finite difference,

$$\left.\frac{d\xi}{dd_1}\right|_{\mathrm{num}} = \frac{\xi(d_1+h) - \xi(d_1-h)}{2h}, \qquad h = 0.25\,\mathrm{mm}, \tag{25}$$

computed independently in the tangential and sagittal planes with the worst-case (larger) value retained. The construction tolerance required to keep ξ within 5% of the target is then

$$|\delta d_1|_{\max} = \frac{0.05\,\xi^*}{|d\xi/dd_1|}, \tag{26}$$

which has units of length and can be compared directly against the positioning accuracy of standard CNC machining ($\sim$25–50 μm) or manual mirror placement ($\sim$100–200 μm). The analytic sensitivity of the round-trip optical path length,

$$\frac{dL_{\mathrm{opt}}}{dd_1} = 1 + \frac{1}{\cos\theta}, \tag{27}$$

is the same for all configurations at fixed θ and is retained for reference; for $\theta = 5^\circ$ this gives ≈ 2.004, meaning a 1 mm shift in d_1 changes the round-trip length by $\approx$ 2 mm and shifts the cavity FSR by $\Delta\nu_{\mathrm{FSR}} \approx c\,\Delta L/L^2$, a useful diagnostic for cavity length servo design.

Stage 6 (transverse-mode degeneracy). Compute the round-trip Gouy phase in both the tangential and sagittal planes for all three cavity states relevant to the design: the empty cavity, the crystal-loaded cavity at 1064 nm, and the crystal-loaded cavity at 532 nm. A candidate is rejected if *any* transverse mode (n, m) with $n + m \leq 5$, in either plane and any of the three states, resonates within one cavity linewidth of the carrier. This stage is necessary because the empty-cavity and loaded-cavity mode spacings differ, so a geometry that is non-degenerate loaded can still be degenerate empty, which would compromise the alignment and diagnostic procedure the empty-cavity stability requirement (Stage 1) is meant to protect.

A geometry surviving all six stages is guaranteed to hit the optical targets and to remain alignable without the crystal; it is not, on its own, guaranteed to be mechanically buildable, which is why the clipping check of Section 6.1 is applied as an additional, currently manual, constraint on top of the six-stage computational pipeline.

6. RESULTS

6.1. Mechanical Clipping at the Tombstone Mounts

The six-stage pipeline above evaluates the cavity as an idealised ray-optics problem with point-like mirrors. In practice each mirror sits on a tombstone mount with a finite footprint, and the crossing beams of a bow-tie cavity pass close enough to the adjacent mounts that clipping becomes a real constraint at small radius of curvature, where the diagonal arms must crowd in close to keep d_1 short.

All candidates with small radii of curvature or small half angle, such as $R_c = 23$ mm, pass every computational filter and have favorable astigmatism as low as $\sim$1–3%. However, the small half angle ensures beam will clip even the smallest mounts at the tight d_1 this ROC requires. This is a useful caution for the rest of the design process: a configuration that looks optimal in (R_c, d_1, d_2) space can still not be physically buildable, and the search space should be restricted to mirrors with enough clearance.

Restricting to mirrors available in stock from Union Optics, this leaves $R_c = 34.51$ mm and $R_c = 46.01$ mm as the two viable candidate radii of curvature carried forward through the rest of this section.

6.2. Bowtie Half-Angle and the Empty-Cavity Stability Bottleneck

Across every scan, regardless of crystal length or candidate radius of curvature, the single largest rejection category is Stage 1's empty-cavity instability check — typically 74–84% of all rejected candidates, an order of magnitude larger than any other filter.

For the 10 mm crystal at $\theta = 5°$, $R_c = 34.51$ mm ($\xi_{\text{green}} = 1.0$, $d_1 \in [38, 58]$ mm, $d_2 \in [5, 200]$ mm), the search found 1299 dual-stable configurations out of $\sim 10^6$ candidates scanned, with the rejection breakdown

Rejection cause	Fraction
Empty-cavity unstable	77.8%
Green-mode unstable	10.2%
Green waist out of range	11.5%
Green astigmatism $> 10\%$	0.5%
HOM degeneracy	0.1%

with the target green waist window $w_0 \in [14.97, 21.17]\,\mu$m ($\xi \in [1.0, 2.0]$).

This is the same problem the LIGO design encountered: a geometry that is optically excellent once loaded with the crystal can easily be unstable empty, which is unacceptable since the cavity must be aligned and diagnosed before the crystal is inserted. Practically, this collapses the usable range of d_1 at fixed R_c to a narrow stability island, located close to $d_1 \approx R_c$, since this is roughly where the curved-mirror pair's focal length matches the arm length needed to close the cavity at all.

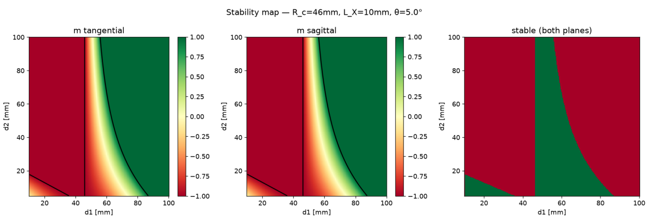


Figure 9: Representative stability map ($l_X = 10$ mm, $R_c = 46$ mm, $\theta = 5.0°$): tangential (left) and sagittal (centre) stability parameter m across (d_1, d_2), and the jointly-stable region (right). The combined stability region collapses to a narrow strip located at $d_1 \approx R_c$, illustrating the empty-cavity bottleneck described above; the same qualitative shape recurs across every crystal length and candidate R_c scanned.

This bottleneck motivated revisiting the bow-tie half-angle θ directly with stability-map data, rather than relying on the LIGO precedent alone. Scanning $\theta \in \{5°, 6°, 7°\}$ for the 20 mm crystal:

- At $\theta = 7°$, neither candidate radius of curvature ($R_c = 34.51$ or 46.01 mm) yields any stable, optically-acceptable configuration, mostly due to high astigmatism.
- At $\theta = 6°$, both candidates recover stable solutions ($R_c = 34.51$ mm over $d_1 \in [34.5, 50]$ mm, $d_2 \in [30, 100]$ mm; $R_c = 46.01$ mm over $d_1 \in [46, 70]$ mm, $d_2 \in [40, 100]$ mm).
- At $\theta = 5°$, both candidates again yield solutions, with the widest stability strips and the most design margin of the three angles tested.

This confirms $\theta = 5°$ as the preferred half-angle for the Berkeley design: not only does it minimise astigmatism, as previously argued, it is also the angle at which the empty-cavity stability bottleneck is least restrictive.

6.3. Stable Configuration Families

Running the six-stage scan at $\theta = 5°$ over the two mechanically-viable radii of curvature identified in Section 6.1 yields stable configuration families for both the $l_X = 10$ mm and $l_X = 20$ mm crystals, summarised in Table 2.

All four configurations sit comfortably inside the Stage-2 waist-matching tolerance, with Δ_{grn} no worse than 2.6% against the $\leq 5\%$ bound. Two trends are consistent across both radii of curvature. First, astigmatism is uniformly worse for the longer crystal: $\mathcal{A}_g$ rises from 5.3% to 6.7% at $R_c = 34.51$ mm and from 7.2% to 9.6% at $R_c = 46.01$ mm when going from $l_X = 10$ to 20 mm, tracking the higher achieved ξ_g ($\sim$1.0 vs. $\sim$1.5–1.6) that comes with the 20 mm crystal's tighter target focus. Second, the construction tolerance on d_1 tightens with crystal length for the same reason: $|\delta d_1|_{\max}$ falls from 88.1 to 68.0 μm at $R_c = 34.51$ mm and from 81.1 to 57.0 μm at $R_c = 46.01$ mm, the same qualitative penalty for operating closer to the Boyd–Kleinman optimum discussed in Section 2.2, now showing up as a mechanical-placement requirement rather than a pump-power one.

Because the signal and pump fields experience different optical path lengths through the crystal ($n_a l_X$ and $n_b l_X$ respectively, with $n_a = 1.830$ and $n_b = 1.889$ at 1064 and 532 nm), the two FSRs are not identical:

$$\text{FSR}_{a,b} = \frac{c}{(d_1 - l_X) + 2d_3 + d_2 + n_{a,b}\, l_X}. \tag{28}$$

The signal-field finesse,

$$\mathcal{F} = \frac{\pi\sqrt{r_{\text{rt}}}}{1 - r_{\text{rt}}}, \qquad r_{\text{rt}} = \sqrt{1 - T_{1,r}}\,\sqrt{1 - T_{2,r}}\, e^{-\alpha l_X/2}, \tag{29}$$

depends only on the round-trip loss and not on the cavity geometry, so it is essentially independent of R_c at fixed crystal length: $\mathcal{F}_r = 38$ for every row in Table 2, with $\mathcal{F}_g$ dropping slightly from 273 to 271 between the 10 and 20 mm crystals as the length-dependent crystal-loss term $e^{-\alpha l_X/2}$ grows. The corresponding signal linewidth and OPA squeezing bandwidth are

$$\Delta\nu = \frac{\text{FSR}_r}{\mathcal{F}}, \qquad \nu_{\text{OPA}} = \frac{\Delta\nu}{2} = \frac{\kappa}{2\pi}, \tag{30}$$

where κ is the total signal decay rate entering the Hall input–output formalism. Bandwidth is set almost entirely by L_{opt} through FSR_r rather than by crystal

Table 2: Representative stable configurations at $\theta = 5°$ for both crystal lengths, restricted to the two mechanically-viable radii of curvature identified in Section 6.1.

		Geometry				*Green (532 nm)*			*Red (1064 nm)*		*Cavity performance*				*Buildability*	
l_X [mm]	R_c [mm]	d_1 [mm]	d_2 [mm]	d_3 [mm]	$L_{\rm opt}$ [cm]	$w_{0,g}$ [μm]	ξ_g	$\mathcal{A}_g$ [%]	$w_{0,r}$ [μm]	$\mathcal{A}_r$ [%]	FSR$_r$ [MHz]	FSR$_g$ [MHz]	$\Delta\nu$ [MHz]	$\nu_{\rm OPA}$ [MHz]	$\lvert d\xi/dd_1\rvert$ [m^{-1}]	$\lvert\delta d_1\rvert_{\max}$ [μm]
10	34.51	42.71	64.73	53.92	22.36	20.96	1.02	5.3	30.28	4.2	1341	1337	35.3	17.6	579.1	88.1
10	46.01	54.21	114.31	84.58	34.62	21.15	1.00	7.2	30.56	5.8	866	865	22.8	11.4	617.7	81.1
20	34.51	49.78	28.84	39.46	17.41	24.14	1.54	6.7	36.52	4.4	1722	1710	45.4	22.7	1130.9	68.0
20	46.01	61.24	57.36	59.53	25.43	23.81	1.58	9.6	36.20	6.3	1179	1174	31.1	15.5	1388.0	57.0

length directly: the most compact configuration ($R_c = 34.51$ mm, $l_X = 20$ mm, $L_{\rm opt} = 17.4$ cm) gives the widest bandwidth at 22.7 MHz, while the longest path ($R_c = 46.01$ mm, $l_X = 10$ mm, $L_{\rm opt} = 34.6$ cm) gives the narrowest at 11.4 MHz.

7. DESIGN RECOMMENDATION

The crystal and mirror procurement now spans two crystal lengths and two radii of curvature: $l_X \in \{10, 20\}$ mm and $R_c \in \{34.51, 46.01\}$ mm, the two ROC values established as mechanically viable in Section 6.1. Rather than a single preferred operating point, the Berkeley source is being built and validated as four parallel candidate geometries, summarised in Table 2. The following section will discuss these chosen configurations in the context of HOM analysis (Section 7.2), and thermal stability (Section 7.3).

7.1. LIGO Design Comparison

One configuration in particular is worth calling out before the full comparison: the $l_X = 10$ mm, $R_c = 46.01$ mm build sits very close to the aLIGO VOPO cavity [20], the design this entire program is descended from. Table 3 compares the two directly; the mirror parameters reflect the updated 2025 procurement specification (E2500276-V2).

Table 3: Berkeley's $l_X = 10$ mm, $R_c = 46.01$ mm candidate vs. the LIGO VOPO final design [20] (mirror spec E2500276-V2, 2025).

Parameter	Berkeley (10/46.01)	LIGO VOPO
l_X	10.0 mm	10.505 mm
ROC, curved mirrors	46.01 mm	50.0 mm
Bowtie half-angle θ	5.0°	6.0°
$w_{0,\rm green}$ at crystal	21.15 μm	21.1 μm
$w_{0,\rm red}$ at crystal	30.56 μm	30.0/29.9 μm
Astigmatism (green/red)	7.2% / 5.8%	∼9% / ∼10%
FSR (green/red)	865 / 866 MHz	851.9 / 853.5 MHz
Finesse (green/red)	273 / 38	∼131 / ∼43
T_1 (green/red)	2% / 10–15%	2% / 7%
T_2 (green/red)	0.26% / 0.26%	0.1–0.25% / 0.1%

The agreement is genuine where it matters most for the nonlinear interaction: the green pump waist (21.15 vs. 21.1 μm) is essentially identical across the two designs. The LIGO M1 handles the 532 nm pump ($T_{1,g} = 2\%$, matching Berkeley exactly), while M2 handles the 1064 nm signal ($T_{1,r} = 7\%$, less than our 10–15% design range). The earlier LIGO generation used a single dichroic coupler at equal reflectivity for both colours; that symmetry is gone.

Given that all other parameters are comparable, the $l_X = 10$mm, $R_c = 46.01$mm configuration serves as a direct baseline against the LIGO VOPO, allowing deviations introduced by the remaining three candidate geometries- longer crystal, tighter ROC, or both- to be interpreted as clean departures from a known good design rather than from an untested one.

7.2. Beampath Validation & HOM Spectra

Figure 10 overlays the ABCD-propagated beam profiles for the $l_X = 10$ mm, $R_c = 46.01$ mm candidate against the LIGO VOPO geometry. The two cavities are nearly indistinguishable in both spot size and accumulated Gouy phase, consistent with the near-identical green waists reported in Table 3. The Berkeley cavity is marginally more compact, as expected from the smaller bowtie half-angle (5° vs. 6°).

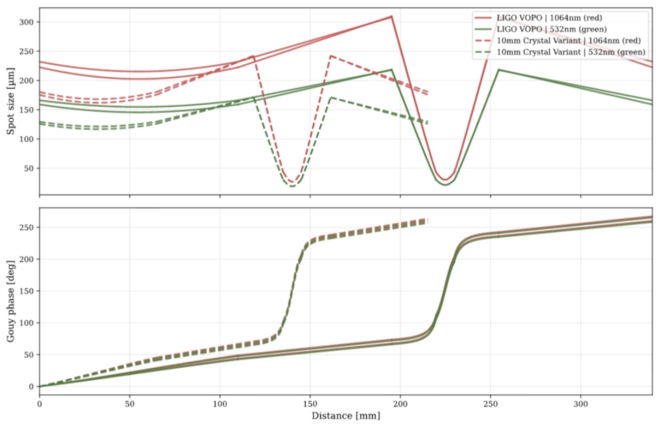


Figure 10: ABCD beam profiles for the $l_X = 10$ mm, $R_c = 46.01$ mm candidate (solid) overlaid on the LIGO VOPO geometry (dashed), at both 532 nm and 1064 nm with sagittal and tangential planes resolved.

Figure 11 shows the HOM resonance combs for the $l_X = 10$ mm, $R_c = 34.51$ mm configuration, computed analyti-

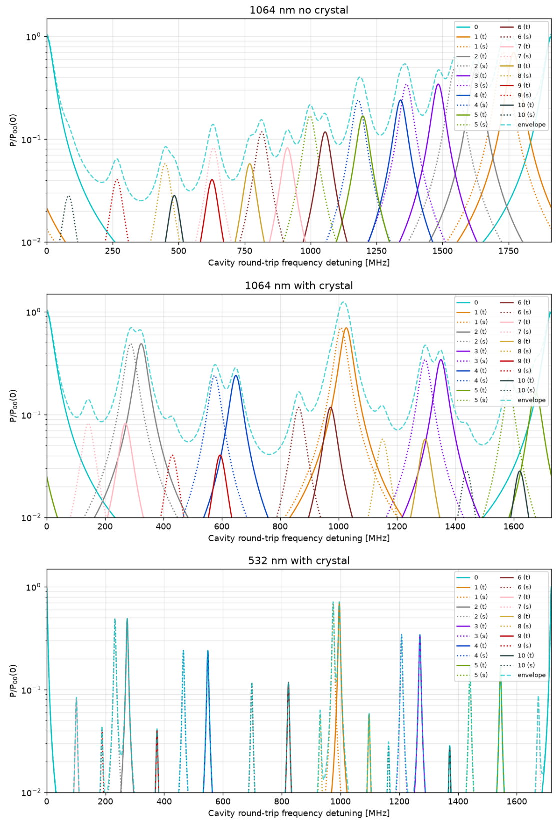


Figure 11: Analytic HOM resonance combs for $l_X = 10\,\mathrm{mm}$, $R_c = 34.51\,\mathrm{mm}$. Loading the crystal separates the higher-order modes from the carrier; the tangential and sagittal resonances of each order are resolved as dashed and dotted curves respectively.

cally from the ABCD-derived round-trip Gouy phase for the empty cavity and each crystal-loaded field. Loading the crystal visibly separates the higher-order modes from the carrier relative to the empty-cavity comb, reproducing the qualitative effect discussed in Section 6.2: the crystal's refractive index breaks the near-degeneracy that the empty-cavity stability requirement otherwise forces the geometry toward. The tangential/sagittal splitting within each mode order reflects the cavity astigmatism; the $R_c = 46.01\,\mathrm{mm}$ configurations produce wider splitting, consistent with their larger astigmatism in Table 2. At $R_c = 46.01\,\mathrm{mm}$ orders 5 and 10 approach co-resonance in the tangential plane at 1064 nm; at $R_c = 34.51\,\mathrm{mm}$ this shifts to a near-degeneracy of order 7 in the green. Neither is disqualifying, but the dominant HOM risk shifts wavelength as ROC changes, a distinction invisible to the stability analysis alone.

7.3. Thermal Self-Consistency

One further question the six-stage filter does not answer: does the cold-cavity eigenmode remain valid once the crystal is heated by the circulating pump?

At finite crystal absorption α, the circulating green pump deposits power $P_{\mathrm{abs}} = P_{\mathrm{circ}}(1 - e^{-\alpha l_X})$ in the ppKTP crystal, inducing a thermal lens with focal length

$$f_{\mathrm{th}} = \frac{\pi w_0^2 \kappa_T}{P_{\mathrm{abs}}\,(dn/dT)\,l_X}, \tag{31}$$

where $\kappa_T = 2.0\,\mathrm{W\,m^{-1}\,K^{-1}}$ is the KTP thermal conductivity [23] and $dn/dT = 2.42 \times 10^{-5}\,\mathrm{K^{-1}}$ is the thermo-optic coefficient at 532 nm. This lens can in principle be inserted as a thin-lens ABCD element $M_{\mathrm{th}} = \begin{bmatrix} 1 & 0 \\ -1/f_{\mathrm{th}} & 1 \end{bmatrix}$ at the crystal centre, shifting ξ_{eff} away from the cold-cavity prediction.

The natural dimensionless measure of thermal-lens strength is f_{th}/d_1: when $f_{\mathrm{th}} \gg d_1$, the lens is negligible relative to the focusing power of the curved mirrors ($\sim 2/R_c$) and the cold-cavity design is self-consistent. Figure 12 (left) plots this ratio for all four candidate geometries as a function of circulating pump power. The peak intensity at the crystal face,

$$I_{\mathrm{peak}} = \frac{2P_{\mathrm{circ}}}{\pi w_0^2}, \tag{32}$$

is shown alongside the grey-tracking damage threshold in Figure 12 (right).

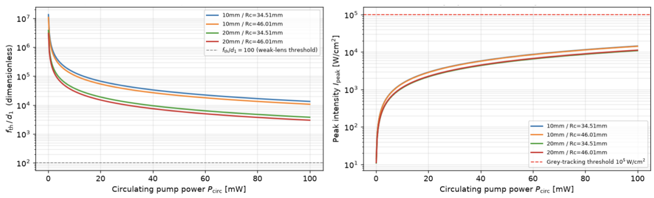


Figure 12: *Left:* Ratio f_{th}/d_1 versus circulating pump power for the four candidate geometries. Values far above unity confirm that the thermal lens is negligible relative to the cavity geometry. *Right:* Peak intensity I_{peak} at the crystal face versus circulating pump power. The red dashed line marks the grey-tracking damage threshold $\lesssim 10^5\,\mathrm{W\,cm^{-2}}$ reported for ppKTP [23, 22].

Figure 12 confirms $f_{\mathrm{th}}/d_1 \gg 1$ at all relevant operating powers, so the cold-cavity ABCD eigenmode is self-consistent: thermal lensing does not measurably perturb ξ_{eff}. Second, I_{peak} remains well below the grey-tracking damage threshold, confirming the ability to operate closer to the Boyd–Kleinman optimum $\xi^* = 2.84$ with the longer crystal.

7.4. Discussion

The four candidate geometries trade off against one another in the same three ways established analytically in Table 2: $R_c = 34.51\,\mathrm{mm}$ gives a more compact cavity, wider OPA bandwidth, lower astigmatism, and a looser construction tolerance than $R_c = 46.01\,\mathrm{mm}$ at fixed crystal length; and $l_X = 10\,\mathrm{mm}$ gives lower astigmatism and a looser tolerance than $l_X = 20\,\mathrm{mm}$ at fixed ROC, at the cost of the higher threshold pump power quantified in Section 4. The HOM analysis adds a further nuance: the dominant co-resonance risk shifts from the red field to the green as ROC decreases, a distinction invisible to the stability analysis alone but potentially relevant to the squeezing floor. No candidate is eliminated by either analysis, so the choice between them remains a question of which trade-off is weighted more heavily once the squeezing budget of Section 4 is folded in.

[1]Correspondence: katiegray@berkeley.edu

REFERENCES

[1] C. M. Caves, *Quantum-mechanical noise in an interferometer*, Phys. Rev. D **23**, 1693–1708 (1981). doi:10.1103/PhysRevD.23.1693

[2] M. J. Collett and C. W. Gardiner, *Squeezing of intracavity and travelling-wave light fields produced in parametric amplification*, Phys. Rev. A **30**, 1386–1391 (1984). doi:10.1103/PhysRevA.30.1386

[3] C. W. Gardiner and M. J. Collett, *Input and output in damped quantum systems: quantum stochastic differential equations and the master equation*, Phys. Rev. A **31**, 3761–3774 (1985). doi:10.1103/PhysRevA.31.3761

[4] G. J. Milburn and D. F. Walls, *Production of squeezed states in a degenerate parametric amplifier*, Opt. Commun. **39**, 401–404 (1981). doi:10.1016/0030-4018(81)90232-7

[5] C. W. Gardiner and P. Zoller, *Quantum Noise: A Handbook of Markovian and Non-Markovian Quantum Stochastic Methods with Applications to Quantum Optics*, 3rd ed., Springer, Berlin (2004). ISBN 978-3-540-22301-6.

[6] R. E. Slusher, L. W. Hollberg, B. Yurke, J. C. Mertz, and J. F. Valley, *Observation of squeezed states generated by four-wave mixing in an optical cavity*, Phys. Rev. Lett. **55**, 2409–2412 (1985). doi:10.1103/PhysRevLett.55.2409

[7] L.-A. Wu, H. J. Kimble, J. L. Hall, and H. Wu, *Generation of squeezed states by parametric down conversion*, Phys. Rev. Lett. **57**, 2520–2523 (1986). doi:10.1103/PhysRevLett.57.2520

[8] H. Vahlbruch, M. Mehmet, S. Chelkowski, B. Hage, A. Franzen, N. Lastzka, S. Goßler, K. Danzmann, and R. Schnabel, *Observation of squeezed light with 10-dB quantum-noise reduction*, Phys. Rev. Lett. **100**, 033602 (2008). doi:10.1103/PhysRevLett.100.033602

[9] H. Vahlbruch, M. Mehmet, K. Danzmann, and R. Schnabel, *Detection of 15 dB squeezed states of light and their application for the absolute calibration of photoelectric quantum efficiency*, Phys. Rev. Lett. **117**, 110801 (2016). doi:10.1103/PhysRevLett.117.110801

[10] M. Mehmet and H. Vahlbruch, *High-efficiency squeezed light generation for gravitational wave detectors*, Class. Quantum Grav. **36**, 015014 (2019). doi:10.1088/1361-6382/aaf448

[11] M. Mehmet, S. Ast, T. Eberle, S. Steinlechner, H. Vahlbruch, and R. Schnabel, *Squeezed light at 1550 nm with a quantum noise reduction of 12.3 dB*, Opt. Express **19**, 25763–25772 (2011). doi:10.1364/OE.19.025763

[12] M. Tse *et al.*, *Quantum-enhanced Advanced LIGO detectors in the era of gravitational-wave astronomy*, Phys. Rev. Lett. **123**, 231107 (2019). doi:10.1103/PhysRevLett.123.231107

[13] J. Lough *et al.*, *First demonstration of 6 dB quantum noise reduction in a kilometre-scale gravitational wave observatory*, Phys. Rev. Lett. **126**, 041102 (2021). doi:10.1103/PhysRevLett.126.041102

[14] R. Schnabel, *Squeezed states of light and their applications in laser interferometers*, Phys. Rep. **684**, 1–51 (2017). doi:10.1016/j.physrep.2017.04.001

[15] M. S. Stefszky, *Low-noise squeezed light for gravitational wave detection*, Ph.D. thesis, Australian National University (2012). ANU Open Research Repository

[16] S. Ast, M. Mehmet, and R. Schnabel, *High-bandwidth squeezed light at 1550 nm from a compact monolithic PPKTP cavity*, Opt. Express **21**, 13572–13579 (2013). doi:10.1364/OE.21.013572

[17] E. D. Hall, *Precision dynamics of resonantly enhanced optical parametric amplifiers*, arXiv:2501.xxxxx (2025).

[18] N. Kijbunchoo, T. McRae, S. Chua, P. Veitch, and D. Sigg, *PPKTP crystal degradation study: preliminary results*, LIGO Technical Note T2500346-v1 (2025).

[19] G. D. Boyd and D. A. Kleinman, *Parametric interaction of focused Gaussian light beams*, J. Appl. Phys. **39**, 3597–3639 (1968). doi:10.1063/1.1656831

[20] J. Miller, L. Barsotti, F. Matichard, M. Evans, A. Fernández, and P. Fritschel, *aLIGO, SQZ, VOPO Cavity, Final Optical Design*, LIGO Technical Note LIGO-TT1700104-v2 (2017).

[21] S. Emanueli and A. Arie, *Temperature-dependent dispersion equations for $KTiOPO_4$ and $KTiOAsO_4$*, Appl. Opt. **42**, 6661–6665 (2003).

[22] K. Goda, *Development of Techniques for Quantum-Enhanced Laser-Interferometric Gravitational-Wave Detectors*, Ph.D. thesis, Massachusetts Institute of Technology, Cambridge (2007).

[23] R. Blachman, P. F. Bordui, and M. M. Fejer, *Laser-induced photochromic damage in potassium titanyl phosphate*, Appl. Phys. Lett. **64**, 1318 (1994).

[24] J. K. Tyminski, *Photorefractive damage in KTP used as second-harmonic generator*, J. Appl. Phys. **70**, 5570 (1991).